
team1
BUILDERS
BRUNCH

Hi 👋 I'm

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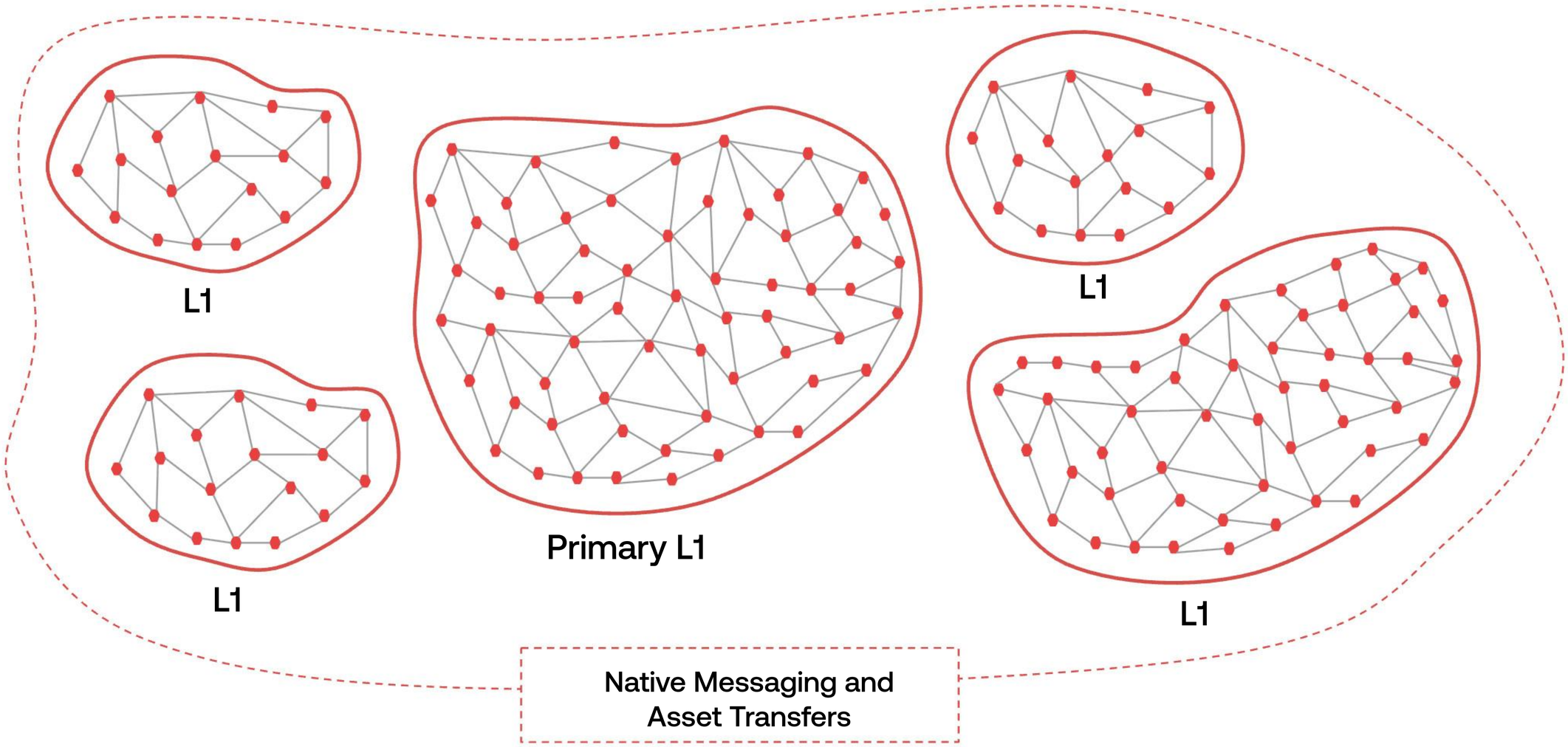


avalanche

Inter Chain Messaging (ICM) between L1s

Intermediate Level

Interoperable network of layer 1s.



Primary network. Three chains.

X-chain

Assets

Native assets with rules baked in equity, bonds, "can't trade until Tuesday."
Relevant to Wallets and exchanges.

C-chain

Contracts

A real EVM Solidity, Geth APIs, Hardhat, Foundry and other toolings.
Ship your existing Ethereum codebase.

P-chain

Coordination

Validators, staking, and the registry of every L1 on the network.
Custom chains get registered here.

06

Two islands.

Two L1s on one laptop. Neither can see the other.

chainA	
EVM chain ID	1001
gas token	AAA
validators	its own
RPC	127.0.0.1:9654
blockchain ID	0xec0b…432f

the P-Chain tx that created it — new on every reset

?

chainB	
EVM chain ID	1002
gas token	BBB
validators	its own
RPC	127.0.0.1:9656
blockchain ID	0xfd85…d64a

the P-Chain tx that created it — new on every reset

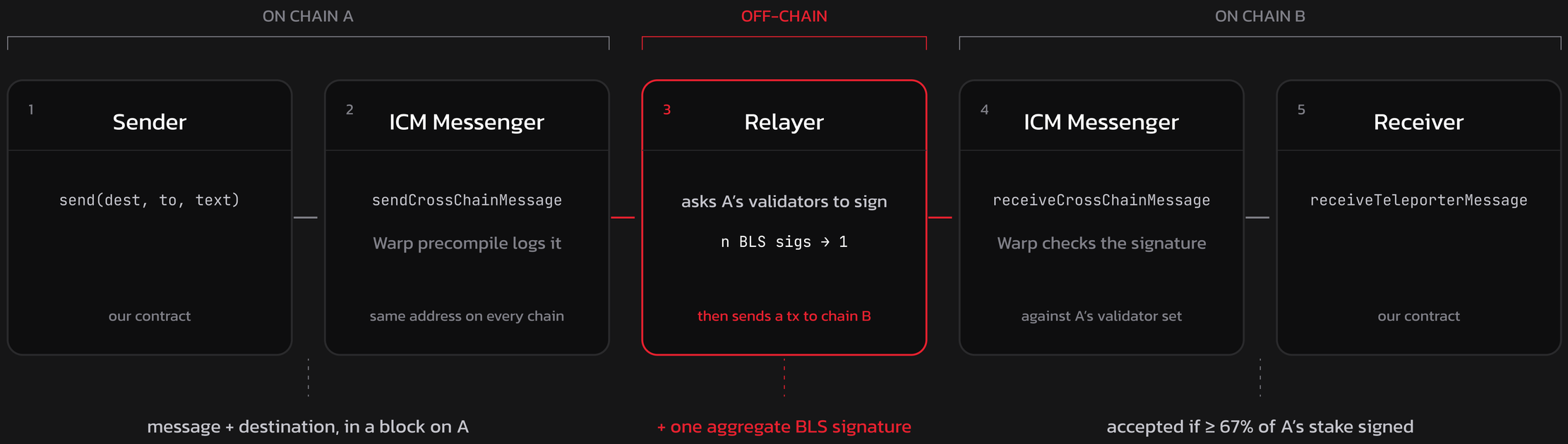
THE QUESTION

How does chain B believe something chain A says?

Each L1 trusts only its own validators.

How a message crosses.

Five hops. Two of them are ours.



The relayer carries it. The signatures make it true.

Who you trust.

Not the relayer. That's the point.

THE TRUST MODEL

Chain B believes chain A's validators. Nobody else.

THE COURIER

Relayer

- ✓ carry the message
- ✓ choose when
- ✗ change a byte
- ✗ invent a message

Anyone can run one.

today it's a process on this laptop

signature

THE WITNESSES

Chain A's validators

each signs the Warp message

with its BLS key

`≥ 67% of stake`

aggregated into one signature

their BLS keys

THE REGISTER

P-Chain

records every L1's

validators and BLS keys

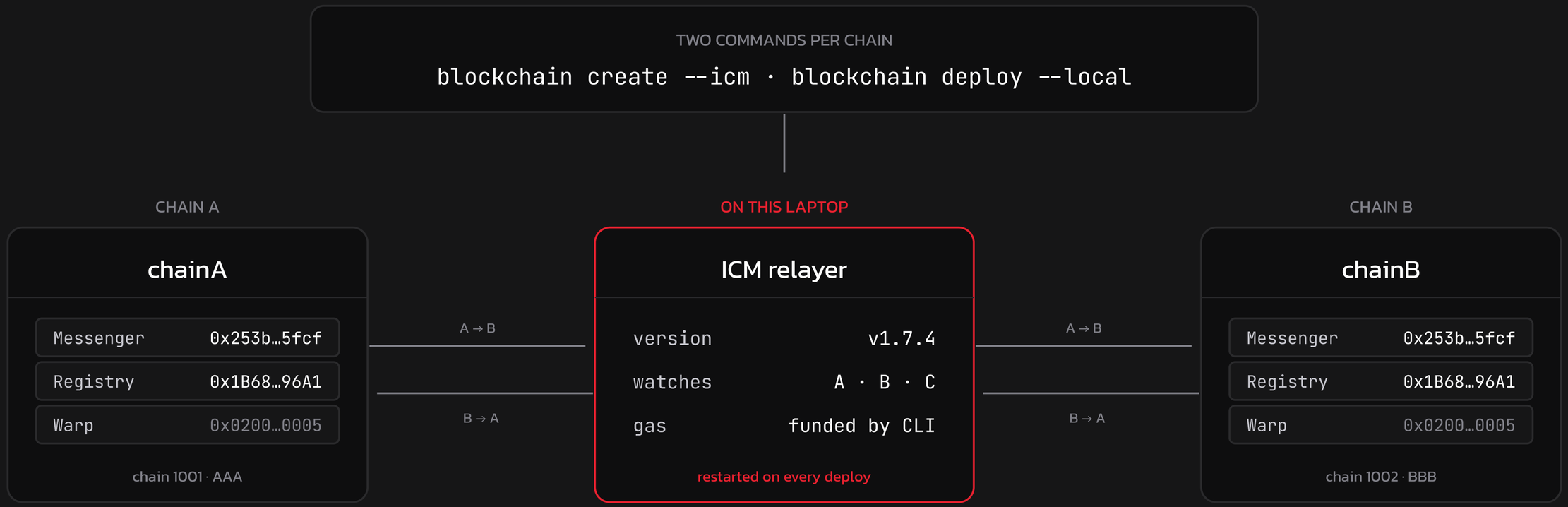
`chain B reads it`

to check A's signature

Anyone can relay. Nobody can forge.

What the CLI set up.

ICM comes pre-installed. We only write the two ends.



Same Messenger address on every chain — it has to be.

a Messenger only accepts messages sent by its own address · deployed with Nick's method

Messenger on both chains. A relayer in between.

Two contracts.

One sends through the Messenger. One only listens to it.

Sender · on chain A

```
function send(bytes32 dest, address to,
              string calldata text) {
  messenger.sendCrossChainMessage(
    TeleporterMessageInput({
      destinationBlockchainID: dest,
      destinationAddress: to,
      feeInfo: TeleporterFeeInfo(address(0), 0),
      requiredGasLimit: 300_000,
      allowedRelayerAddresses: new address[](0),
      message: abi.encode(text)
    }));
}
```

gas chain B must give the Receiver

ICM

Receiver · on chain B

```
function receiveTeleporterMessage(
  bytes32 sourceBlockchainID,
  address originSenderAddress,
  bytes calldata message
) external {
  if (msg.sender ≠ messenger)
      revert OnlyMessenger();

  lastMessage =
      abi.decode(message, (string));
  received++;
}
```

Warp verified it · the Messenger checked origin and replay

Send through the Messenger. Only trust the Messenger.

Only the Messenger knocks.

What stops a fake delivery.

LIVE · just forge-delivery

Call the Receiver directly and claim to be chain A.

THE REAL PATH

Via the relayer

caller	Messenger
signature	checked
result	stored

"gm chain B, love chain A"

A FORGED CALL

Straight to Receiver

caller	0x8db9…52FC
signature	none
result	OnlyMessenger

reverted · nothing stored

WHAT PRODUCTION ADDS

Check the sender

```
sourceBlockchainID
    == chainA
originSenderAddress
    == Sender
```

Ours skips this, on purpose.

any contract on any chain can message it

Check the Messenger. Then check who sent it.

Where this goes.

Messages first. Tokens ride on them.

Today we

- ✓ booted two L1s with ICM
- ✓ sent a message both ways from the CLI
- ✓ wrote our own Sender and Receiver
- ✓ watched a forged delivery revert

Before mainnet

- — check source chain and sender
- — pay relayers with feeInfo
- — find the Messenger via the Registry
- — run, or pick, your relayers

ICM MESSAGES

any bytes, chain to chain

today

ICTT

move AAA onto chain B

interchain tokenTransferrer

FUJI

real validators, real relayers

next

Two L1s. One message. About two seconds.

demo

Questions?

Thank you!



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